Joey Jaehyeok Kim

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Education

Georgia Institute of Technology
M.S. Mechanical Engineering — Aug 2026 – May 2027
B.S. Mechanical Engineering | GPA: 3.78/4.00 — Jun 2022 – Jul 2026

Experience

Tesla — Lathrop, California
Test Engineer — Jan 2025 – Jul 2025

- 98% FPY and 10x mating cycles — Led concept-to-deployment of pogo-pin connectors with ASME Y14.5 GD&T stack-up
- 7% production increase — Led full-cycle development for robotic end effectors for low-voltage testing
- $20,000 cost reduction — Designed fixtures for critical electrical equipment using CAD and DFM principles
- 4% EoL FPY increase — Integrated DFMEA and corrective actions across Maintenance and Production
- 15min/week downtime reduction — Designed mobile testing station to troubleshoot components off the line
- 30min/week downtime reduction — Streamlined component pipeline with Supply Chain via demand forecasting
- Diagnosed electronic subassembly failures via DFMEA, fault injection, and data analysis

Cognosos — Atlanta, Georgia
Mechanical Engineer — Jan 2024 – Aug 2024

- 97% FPY across 300+ Real Time Location System (RTLS) units — Assembled and tested for verified 1-year operation
- 6% vendor acceptance increase — Conducted capability audits, and applied DFM to injection molded enclosure CAD
- 5% prototyping turnaround improvement — Implemented a system resolving resource and product release conflicts
- Migrated full SolidWorks library to Onshape with ASME Y14.5-compliant schematic documentation
- Led impact and compression testing, defining pass/fail criteria for all RTLS assemblies

Computational Design and Metamaterials Lab — Atlanta, Georgia
Research Assistant under Dr. Bolei Deng — Aug 2024 – Present

- Developed a differentiable simulation pipeline to optimize novel Triply Periodic Minimal Surface (TPMS) geometries for energy absorption via nonlinear FEM, validated through shear and axial testing
- Designed a bistable tube with dilative and compressive properties as an alternative to surgical endotracheal tubes
- Modeled bistable unit cells to create metamaterials with 2 DoF on flat surfaces and neutral buoyancy using CAD and FEA

Projects

Helical Compression Spring Design Optimization — Mar 2026 – Apr 2026

- Optimized a helical compression spring for minimum mass formulating 3 design variables against 7 engineering constraints and validating global optimality via multi-start sampling
- 10% stress constraint relaxation yielded 18% mass reduction, as sensitivity analysis identified allowable shear stress as the dominant design driver

2D Finite Difference Thermal Simulation of CPU/GPU Chips — Jan 2026 – Mar 2026

- Modeled heating, steady-state, and cool-down phases across a 3-layer chip stack (silicon, thermal paste, copper spreader) with localized heat sources and convective cooling to predict peak junction temperatures and characterize thermal behavior
- $<1\%$ error across 3 cross-validated finite difference solvers confirming thermal paste as the bottleneck to heat dissipation

Automated Faulty PCB Detection, Thermal Imaging — May 2025 – Jun 2025

- Generated synthetic thermal image datasets using a diffusion model to simulate rare failure modes and validate detection
- 98% classification accuracy on thermal defect detection pipeline using Gaussian smoothing, morphological filtering, and region-based segmentation to isolate abnormal heat signatures on PCBs

TPMS for Headgear Impact Protection — Sep 2025 – Present

- 7% greater energy absorption over EPS foam — Validated through FEA comparisons between conventional EPS foam and TPMS lattices
- Simulated heat transfer and CFD through TPMS models to analyze airflow and temperature at transient and steady states
- Maintained HIC < 250 and BrIC < 0.45, meeting DOT impact requirements through iterative tuning of lattice parameters

Voice-Controlled Robotic Arm Assistant — Jul 2025 – Present

- Designed and prototyped a 6-DOF robotic arm using CAD with integrated AI voice assistant
- Developed motion control system using inverse kinematics, logging, and unit tests for reliability
- Performed FEA stress analysis and fatigue life estimation on arm linkages to validate structural integrity
- Integrated DC motors, IR sensors, solenoids, potentiometers, and pneumatic actuators for multi-axis actuation

Skills

- **Design:** SOLIDWORKS, CATIA, Onshape, AutoCAD, GD&T (ASME Y14.5), DFM, DFMEA, tolerance stack-up
- **Simulation & Analysis:** FEA, finite difference methods, CFD, fatigue analysis, thermal modeling
- **Prototyping & Fabrication:** 3D printing, soldering, laser cutting, CNC lathe, injection molding
- **Test & Validation:** EVT/DVT/PVT process, NPI, RCA, Vector CANape, LabVIEW, Siemens HMI
- **Programming:** Python, MATLAB, C++, Java, Go, Swift, TP Programming